NeuroSymbolic AI Assignment 2026/2027: Card Suit Game with DeepProbLog

Problem Description

In this assignment, two players each play a card. The outcome of the game depends *only on the suit* of each card, not on its rank. You will use DeepProbLog to determine the outcome from two card images.

Game Rules

The game uses exactly three suits: **Hearts**, **Clubs**, and **Spades**. The following rules determine which suit beats which:

Hearts beats **Clubs**	**Clubs** beats **Spades**
Spades beats **Hearts**	**Same suit**: **Draw**

If the first player's suit beats the second player's suit, Player 1 wins; if the second player's suit beats the first player's suit, Player 2 wins. Two cards of the same suit always produce a draw.

Objective

Design and train two systems to predict the game outcome (*Player 1 Wins*, *Player 2 Wins*, or *Draw*) from a pair of card images:

1. **Approach A (DeepProbLog):** a neuro-symbolic model that combines visual recognition with logical game rules.
2. **Approach B (Baseline):** a purely neural model that predicts the outcome directly from both images.

Assignment Tasks

1. **Dataset Preparation**
   - Use the *Cards Image Dataset–Classification* from Kaggle: https://www.kaggle.com/datasets/gpiosenka/cards-image-datasetclassification
   - Keep only cards belonging to Hearts, Clubs, and Spades; exclude Diamonds and the Joker. Ignore card ranks when computing game outcomes.
   - Construct a dataset of pairs $(I_1, I_2, \text{outcome})$ and label each pair using the game rules.
2. **Approach A: The DeepProbLog Model**
   - Define a neural predicate that maps each card image to a probability distribution over $\{\text{hearts}, \text{clubs}, \text{spades}\}$.
   - Combine the neural predicate with your ASP rules to infer the outcome.

3. **Approach B: The Neural Baseline**
    - Implement a standard neural model that takes both card images as input and predicts the final outcome directly.
4. **Experiments & Comparison**
    - Train both models and evaluate accuracy and F1 score on the test set.
    - Compare training time and computational efficiency.
    - Repeat experiments using 20%, 50%, and 100% of the training examples, keeping validation and test data unchanged.
    - Discuss the results and differences between the two approaches.
5. **Presentation**
    - Prepare an 8–12 slide deck summarizing DeepProbLog theory, the implemented methods, experimental results, and your comparison.

Submission Deadline

The deadline for submitting the project is **18 December**. By this date, students must submit the presentation slides and the source code. Moreover, each group will present their work during the same day.